

Closing the Output-Only Inverse Loop

Sunday 27 September 2026: finite-record information (U1), segment-length refinement and twin inversion (U2, Gate G3)

Technical writeup, continued

This note completes the weekend's output-only chain. Saturday showed that complex transmissibilities can be estimated from finite stochastic multi-sensor records and characterised their bias and covariance (Gate G2). Sunday turned that measurement description into parameter information, found that the remaining Welch bias was parameter-significant at the Saturday setting, fixed it by changing the segment length, and then inverted finite twin records for (β, R) from a distant starting point, with an uncertainty estimate formed from the record alone. The target throughout is

$$Y(t) \longrightarrow \hat{T}(\omega) \longrightarrow (\hat{\beta}, \hat{R}), \quad \gamma \text{ fixed.} \quad (1)$$

SSI remains a comparison branch only. Colour convention as before: findings in green, caveats in red, definitions and theory in blue.

What this note does not establish. Field validity of the EMM; robustness to directional or multiple independent wave forcing; coloured or cross-correlated sensor noise; a mis-specified noise PSD; generality across sea states other than the one JONSWAP spectrum used. All results are *synthetic twin* results (Section 7).

1 Setting and notation

Parameters and features. $\theta = [\ln \beta, \ln R]^T$, $\gamma = \gamma_0$ fixed, $p_0 = (4.6985 \times 10^{-5}, 1.4548 \times 10^{-3}, 0.3830)$. Four sensors (Level 2C: s2, s6, s14, s26 at $r/R_0 = 0.3, 0.3, 0.5, 0.9$ on the incidence axis), reference s26. At each retained frequency the feature block is

$$z_k = [\text{Re } T_1, \text{Re } T_2, \text{Re } T_3, \text{Im } T_1, \text{Im } T_2, \text{Im } T_3]_{\omega_k}^T, \quad T_j = H_j/H_r, \quad (2)$$

stacked frequency-major into $f_T \in \mathbb{R}^{6n_F}$. The model map $f_T(\theta)$ is the EMM point ratio (S1); the data map is the corrected Welch estimate (S2), $\hat{T}_j = \hat{S}_{jr}/(\hat{S}_{rr} - S_{nn})$.

Local information. With block-diagonal covariance $\Sigma_T = \text{blkdiag}(\Sigma_k)$ and Jacobian $J = \partial f_T / \partial \theta$ (blocks $J_k \in \mathbb{R}^{6 \times 2}$),

$$F = J^T \Sigma_T^{-1} J = \sum_k J_k^T \Sigma_k^{-1} J_k, \quad \text{Cov}(\hat{\theta}) \approx F^{-1}, \quad \sigma_i = \sqrt{(F^{-1})_{ii}}. \quad (3)$$

A systematic feature error b maps into parameter space as

$$\delta \theta_{\text{bias}} = F^{-1} J^T \Sigma_T^{-1} b, \quad d_{\text{bias}} = \sqrt{\delta \theta_{\text{bias}}^T F \delta \theta_{\text{bias}}}, \quad (4)$$

the bias in units of the uncertainty ellipsoid. Per-bin bias against per-bin scatter is the wrong test: random scatter averages down across bins, a systematic error does not.

Welch-expected bias (Saturday). A Hann segment of length L measures a spectral-window average,

$$T_{\text{pred},j}(\omega_k) = \frac{\sum_q K_L(\omega_k - \omega_q) G_q H_j(\omega_q) H_r^*(\omega_q)}{\sum_q K_L(\omega_k - \omega_q) G_q |H_r(\omega_q)|^2} \quad (5)$$

(correction on, so no S_{nn} term), with K_L the normalised Hann kernel and G_q the input PSD. $b_{\text{pred}} = f(T_{\text{pred}}) - f_T(\theta)$ has no Monte Carlo noise; it uses the twin's known JONSWAP spectrum, so it is a diagnostic of this experiment, not a correction available output-only.

2 U1: finite-record information at the Saturday setting

`runTransmissibilityCovariance.m` generated 200 fresh records at the Saturday production configuration ($N = 65536$, 109 min, JONSWAP $H_s = 0.05$ m, $\omega_p = 5$ rad/s, 10% noise stress case, $\sigma_n = 6.67 \times 10^{-3}$ m/s², correction on). Each record was estimated at both $L = 1024$ and $L = 512$, so

all treatments see identical data. The Jacobian was computed by central differences ($h = 10^{-3}$) through S1 on all 84 ramp-free $L = 1024$ bins (420 EMM solves, cached); the $L = 512$ bins and the decimated production bins are subsets of that grid.

Four covariance treatments. A: $L = 512$, 42 bins, block diagonal. B: $L = 1024$, every second bin (42), block diagonal (the Saturday production setting). C: $L = 1024$, all 84 bins, block tridiagonal with the adjacent-bin cross-covariances $C_{k,k+1}$ (shrunk towards block diagonal until positive definite, $a = 0.90$). D: $L = 1024$, all 84 bins, block diagonal (deliberately ignores the lag-1 correlation).

Held-out calibration. The linearised estimator $\hat{\theta} = \theta_0 + F^{-1} J^T \Sigma^{-1} (z - f_0)$ is applied to every draw with 2-fold cross-fitting: Σ_h from one half, applied to the other, and compared with the σ that the same Σ_h claims. The empirical spread is the pooled within-fold standard deviation. With about 200 draws the ratio $\sigma_{\text{emp}}/\sigma_{\text{pred}}$ has a Monte Carlo standard error of about $1/\sqrt{2 \cdot 199} \approx 0.05$.

Treatment	n_F	dim	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	corr	emp/pred	d_{bias} (MC)	d_{pred}
A $L512$, blkdiag	42	252	0.0037	0.0007	0.832	1.38, 1.35	3.68	3.68
B $L1024$ dec., blkdiag	42	252	0.0047	0.0009	0.839	1.10, 1.05	0.96	0.91
C $L1024$ full, lag-1 banded	84	504	0.0042	0.0008	0.840	1.37, 1.23	1.19	1.13
D $L1024$ full, blkdiag	84	504	0.0033	0.0006	0.839	1.51, 1.41	1.38	1.29

Table 1: U1 (transmissibilityCovariance_20260927_124013). σ is for one 109-minute record. Jacobian step check: $\|J_h - J_{h/2}\|/\|J_h\| = 4.2 \times 10^{-6}$, σ changes 5×10^{-6} .

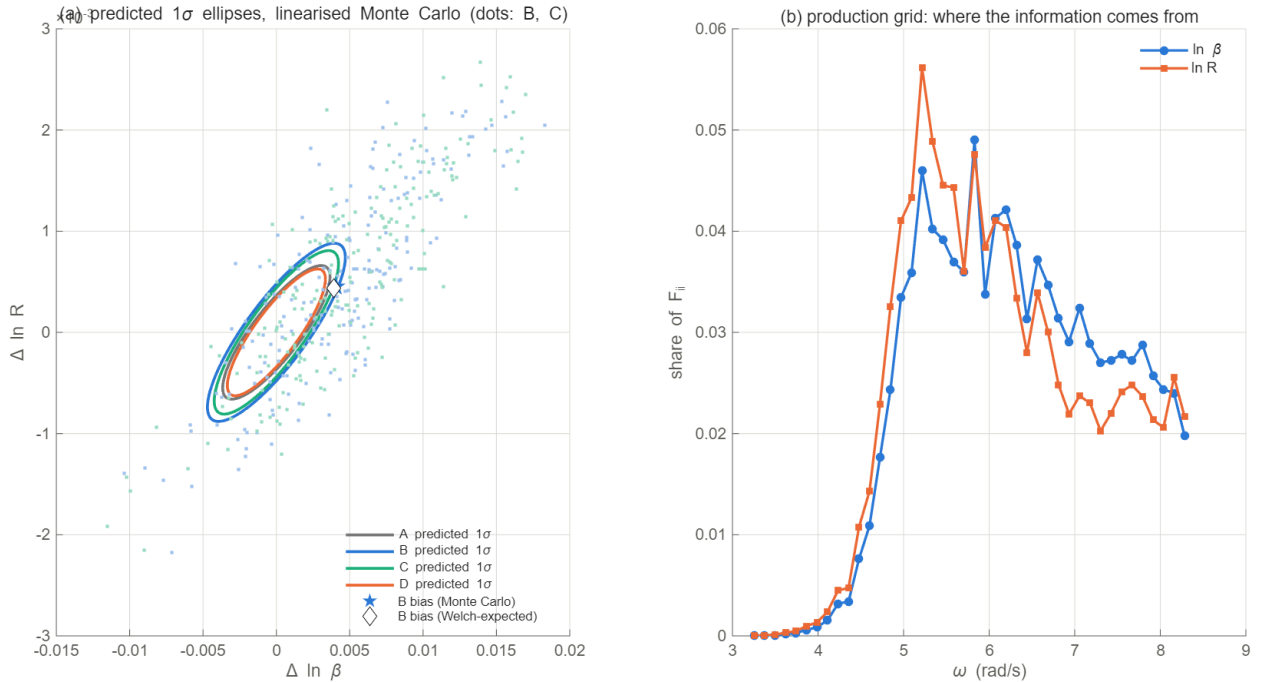


Figure 1: U1. (a) Predicted 1σ ellipses for A to D and the cross-fitted linearised estimates for B and C (dots), with the B parameter bias (Monte Carlo and Welch-expected). (b) Share of the diagonal Fisher information per frequency on the production grid: almost nothing below 4.3 rad/s, most from 4.7 to 6.6 rad/s.

The decimated covariance model works. B's held-out scatter matches its prediction (1.10, 1.05). D, which uses every bin as if independent, is visibly overconfident (1.51, 1.41): ignoring adjacent-bin correlation creates fictitious information, confirming Saturday's decision to thin. Decimation costs about 10% in σ relative to C (1.11, 1.09), although C's own calibration is weaker (1.37, 1.23; fold shrinkage 0.85, 0.80), so that figure is indicative.

But the Welch bias is parameter-significant at $L = 1024$. B has $d_{\text{bias}} = 0.96$ (Monte Carlo) and 0.91 (Welch-expected), with $\delta\theta_{\text{bias}} = [+0.0041, +0.0005]$ and $[+0.0039, +0.0004]$: the two independent estimates agree, so this is the real spectral-window effect, not Monte Carlo noise.

Sub-percent feature bias (+0.39% in β) becomes a one-sigma parameter offset once a long record makes σ small. A and B sit on the same kind of 42-bin grid, and d_{pred} falls from 3.68 to 0.91 when L doubles: the effect is set by the segment length.

3 Segment-length refinement: $L = 2048$

`runL2048BiasVarianceCheck.m` regenerated the same 200 records (same seeds; the $L = 1024$ production σ reproduced U1 exactly) and estimated them at $L = 2048$. Because $\Delta\omega_{2048} = \frac{1}{2}\Delta\omega_{1024}$, every fourth $L = 2048$ bin is exactly a production frequency and every second is exactly one of the 84 $L = 1024$ bins (verified with `ismember`), so U1’s Jacobians were reused and no EMM solves were needed.

Candidate	n_F	N_{eff}	lag-1 ρ	$\sigma_{\ln \beta}$	$\sigma_{\ln R}$	emp/pred	d_{MC}	d_{pred}
B $L = 1024$, every 2nd	42	120.4	0.06	0.00470	0.00088	1.10, 1.05	0.96	0.91
E $L = 2048$, every 4th	42	59.7	0.06	0.00653	0.00125	1.04, 0.98	0.21	0.18
F $L = 2048$, every 2nd	84	59.7	0.06	0.00462	0.00088	1.11, 1.04	0.30	0.26

Table 2: Bias-variance check (L2048BiasVarianceCheck_20260927_130816); lag-1 ρ is between retained neighbours (Monte Carlo floor 0.06). One-sigma and Welch bias in physical terms for F: β 0.46% (+0.11%), R 0.088% (+0.011%).

Segment length L (42 or 84 bins)	512	1024	2048
d_{pred} (Welch-expected parameter bias, in σ)	3.68	0.91	0.26 (F), 0.18 (E)

Table 3: The resolution bias falls by about four per doubling of L .

F removes the dominant systematic error at no precision cost. From B to F, d_{pred} falls $0.91 \rightarrow 0.26$ while σ is unchanged ($0.00470 \rightarrow 0.00462$, $0.00088 \rightarrow 0.00088$). The mechanism: a longer segment narrows the spectral window (less resolution bias), and although N_{eff} halves, the finer grid provides twice as many retained bins that are still decorrelated (lag-1 ρ at the floor), so the information is recovered. E proves the same bias reduction on the original 42 frequencies but pays $\sim 1.4\times$ in σ . F is frozen as the production setting.

Scope of the segment-length choice. The bias is fixed while $\sigma \propto N^{-1/2}$, so $d \propto \sqrt{N}$ (statistical argument); the L^{-2} factor is empirical over $512 \leq L \leq 2048$. As a local guide only,

$$d_{\text{pred}} \approx 0.26 \sqrt{\frac{N}{65536}} \left(\frac{2048}{L} \right)^2. \quad (6)$$

$L = 2048$ is adequate for the present 109-minute JONSWAP twin and must be rechecked if the record length or the sea-state spectrum changes (a narrower or bimodal spectrum steepens the input weighting across the window and increases the bias).

4 Frozen production configuration

Record	$N = 65536$, $\Delta t = 0.1$ s (109 min); one coherent JONSWAP incident field in the twin
Estimator	Welch, periodic Hann, $L = 2048$, 50% overlap, $N_{\text{eff}} = 59.7$; reference s26; correction on
Frequencies	every second $L = 2048$ bin in $[3.178, 8.301]$ rad/s: $n_F = 84$, 3.191 to 8.283 rad/s, spacing 0.0614
Features	$\dim f_T = 504$, frequency-major; model side the point ratio $T_j = H_j/H_r$
Covariance	$\text{blkdiag}(\Sigma_k)$, 6×6 blocks; deployable estimate: pooled block jackknife (Section 6.4)
Sensor noise (twin)	$\sigma_n = 6.67 \times 10^{-3}$ m/s ² , $S_{nn} = 1.416 \times 10^{-6}$ (m/s ²) ² per rad/s, white, independent
Solver model	EMM on a 0.2 rad/s node grid, spline-interpolated; safeguarded modified Gauss–Newton

Table 4: Configuration frozen after Sunday. No further nonlinear Monte Carlo is planned.

5 U2: the inversion procedure

`runTransmissibilityTwinInversion.m` implements the complete output-only inversion. For a given record:

1. Features from the record. Segment-level Welch spectra are formed (identical to S2; verified to reproduce `extractTransmissibility` exactly), averaged, corrected for the reference-noise PSD, and stacked into $z \in \mathbb{R}^{504}$ on the F grid.

2. Model evaluator. For a trial θ , the EMM is solved on nodes 0.2 rad/s apart covering the retained band (about 27 solves), the smooth $H_a/(-\omega^2)$ is spline-interpolated to the 84 frequencies and $f_T(\theta)$ formed; evaluations are memoised by θ . At each truth point the evaluator is checked against direct EMM features: the interpolation error e is projected with Eq. (4), $d_{\text{interp}} = \|F^{-1}J^T\Sigma^{-1}e\|_F$, and must be below 0.1.

3. Weighted least squares.

$$\hat{\theta} = \arg \min_{\theta} \Phi(\theta), \quad \Phi(\theta) = \sum_k r_k(\theta)^T \Sigma_k^{-1} r_k(\theta), \quad r = z - f_T(\theta). \quad (7)$$

4. Safeguarded modified Gauss–Newton, started at $\theta_0 = \ln p_0$:

$$\Delta\theta = (J^T \Sigma^{-1} J)^{-1} J^T \Sigma^{-1} r, \quad \theta \leftarrow \theta + \alpha \Delta\theta, \quad \alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}\} \text{ (first that lowers } \Phi). \quad (8)$$

J is reused between iterations (chord method) and recomputed by central differences only when: no α lowers Φ (if J is already current, the iteration stops at the minimum); an accepted step exceeds 3σ in any component; or 3 iterations pass without convergence. Convergence: every step component below 0.01σ .

5. Uncertainty. $C_\theta = (J^T \Sigma^{-1} J)^{-1}$ at the solution, with Σ either the oracle (known in the twin) or estimated from the record itself (Section 6.4).

6. Calibration statistic. For each estimate $q = (\hat{\theta} - \theta_{\text{true}})^T C_\theta^{-1} (\hat{\theta} - \theta_{\text{true}})$, which is χ_2^2 for an unbiased estimator with correct C_θ : $P(q \leq 2.279) = 0.68$, $P(q \leq 5.991) = 0.95$. With a known offset $\delta\theta_W$ it is non-central with mean $2 + d_W^2$, and 95% coverage of $d \approx 0.25$ is about 0.94. With 200 trials the coverage has a standard error of about 0.015.

Experimental design. Two truth points, both inverted from p_0 : A = $(1.2\beta_0, 0.97R_0)$ and B = $(0.8\beta_0, 1.03R_0)$, $\gamma = \gamma_0$. They are displaced +0.18 and −0.22 in $\ln \beta$ (tens of σ , so nonlinear convergence is tested), in opposite directions in both parameters; the fixed sensors (outermost $0.9R_0$) stay on the floe; the band stays inside the validated EMM range. The sea state and the *absolute* sensor noise are the same at every point. Per truth point, disjoint seeds give 200 calibration records (oracle $\Sigma(\theta_{\text{true}})$, Monte Carlo bias), 200 validation records (linearised coverage tests) and one nonlinear-inversion record; 200 further records at p_0 give $\Sigma(p_0)$.

Three uncertainty treatments. G3a uses the oracle $\Sigma(\theta_{\text{true}})$ (does the inversion behave as Fisher theory predicts?); G3b uses a covariance estimated from each record alone (is the deployable uncertainty honest?); G3c uses $\Sigma(p_0)$ at the truth point (sensitivity only).

6 Gate G3 results

	Truth A ($1.2\beta_0, 0.97R_0$)	Truth B ($0.8\beta_0, 1.03R_0$)
$\sigma(\ln \beta, \ln R)$, oracle	0.00498, 0.00095	0.00431, 0.00082
Model interpolation: max rel. error, d_{interp}	1.7×10^{-5} , 0.000	1.7×10^{-5} , 0.000
<i>G3a, oracle $\Sigma(\theta_{\text{true}})$, 200 validation records</i>		
emp/pred	1.11, 1.10	1.04, 1.06
coverage 68%, 95%	0.590, 0.915	0.645, 0.920
mean q (expected $2 + d_W^2$)	2.35 (2.06)	2.35 (2.06)
<i>Mean parameter offset $[\ln \beta, \ln R]$</i>		
Welch-expected $\delta\theta_W$ (d_W)	[+0.00108, +0.00013] (0.24)	[+0.00093, +0.00009] (0.25)
calibration Monte Carlo	[+0.00106, +0.00011]	[+0.00118, +0.00012]
validation mean	[+0.00114, +0.00015]	[+0.00109, +0.00012]
standard error of the mean	[0.00039, 0.00007]	[0.00032, 0.00006]
<i>G3b, record-derived covariance (31 blocks): 95% coverage; scatter/reported σ</i>		
raw	0.705; 1.63, 1.61	0.705; 1.48, 1.45
raw, Hartlap-type	0.780; 1.43, 1.41	0.790; 1.29, 1.27
pooled ± 2 bins	0.930; 1.16, 1.15	0.920; 1.06, 1.07
pooled, Hartlap-type	0.930 ; 1.13, 1.13	0.925 ; 1.03, 1.04
pooled scatter / oracle scatter	1.02, 1.02	1.01, 1.00
<i>G3c, $\Sigma(p_0)$ at the truth point</i>		
emp/claimed; 95% coverage	1.09, 1.08; 0.950	1.10, 1.11; 0.880
<i>Nonlinear inversion from p_0 (oracle Σ), one record</i>		
iterations, J recomputes, model evaluations, time	4, 2, 13, 349 s	5, 3, 21, 581 s
error / σ	[+1.17, +0.80] (joint $q \approx 1.5$)	[+0.10, -0.06]
error in physical terms	$\beta + 0.58\%$, $R + 0.076\%$	$\beta + 0.04\%$, $R - 0.005\%$
nonlinear minus linearised, / σ	[-0.003, -0.002]	[0.000, 0.000]
χ^2/dof (dof 502)	1.093	1.025
Gate: interpolation, G3.1 to G3.4	all PASS	all PASS

Table 5: Gate G3 (transmissibilityTwinInversion_20260927_143438). Gates: G3.1 nonlinear converged, within 3σ of truth and 0.2σ of the linearised estimate; G3.2 oracle emp/pred in $[0.85, 1.20]$ and 95% coverage in $[0.90, 0.99]$; G3.3 mean offset within 3 s.e. of $\delta\theta_W$; G3.4 pooled Hartlap-type 95% coverage ≥ 0.90 .

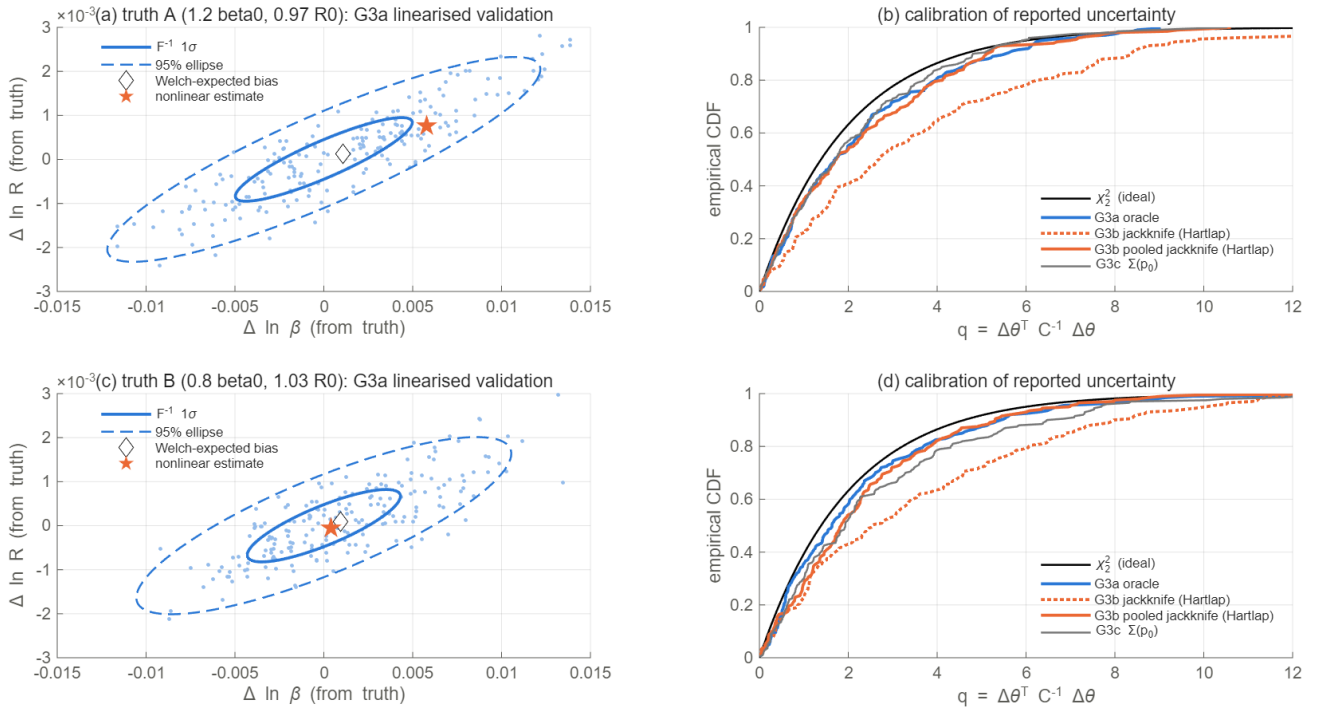


Figure 2: G3. (a, c) Linearised validation estimates (dots) relative to the truth with the oracle $F^{-1} 1\sigma$ and 95% ellipses, the Welch-expected bias (diamond) and the nonlinear estimate (star). (b, d) Empirical CDF of q against χ^2_2 for the oracle, the raw and pooled jackknife (both Hartlap-type) and $\Sigma(p_0)$.

6.1 Nonlinear inversion

The nonlinear solver adds no appreciable error. Starting 18 to 22% away in β , both inversions converged in 4 to 5 iterations, and each landed within 0.003σ of the linearised estimate from the same record: over these excursions the local statistical picture describes the final nonlinear estimate. B needed one more Jacobian update and more evaluations, so the safeguards are used, not redundant. A's componentwise error $[+1.17, +0.80]\sigma$ looks large but lies along the major axis of the ellipse ($\rho \approx 0.84$); its joint $q \approx 1.5$ is inside the 68% region. One record per point says nothing about whether the method is better at A or at B.

6.2 G3a: the oracle Fisher prediction

The local Fisher approximation predicts the Monte Carlo scatter to within about 10%. emp/pred is 1.04 to 1.11 (standard error about 0.05): B is consistent with 1, A about 2 s.e. above. Coverage at 95% is 0.915 and 0.920, but the known offset alone lowers it to about 0.94, so the shortfall is about 1.5 to 2 s.e. Part of the residual optimism is expected from a covariance estimated from 200 records (about 2% in σ from inverting a sample covariance, plus weight noise). Report: good calibration with modest residual optimism, not exact agreement.

6.3 The predicted Welch bias is the observed inversion bias

The mean parameter offset was predicted before the inversion was run. At both truth points the Welch-expected $\delta\theta_W$, the calibration Monte Carlo and the validation mean agree within their standard errors (Table 5):

$$\text{spectral window} \rightarrow \text{bias}(\hat{T}) \rightarrow \delta\theta_{\text{predicted}} \rightarrow \delta\theta_{\text{observed}}. \quad (9)$$

The small positive offset (about +0.1% in β) is not an inverse-model or solver bias; it is the finite-resolution effect identified on Saturday, quantified on Sunday morning and reduced by the choice of L .

6.4 G3b: uncertainty from the record alone

Block jackknife. The record's 63 overlapping Welch segments are grouped into $B = 31$ blocks of 2 consecutive segments (overlap makes neighbouring segments correlated, so blocks, not single segments, are deleted). Deleting block b gives $z_{(-b)}$; per bin

$$\hat{\Sigma}_k = \frac{B-1}{B} \sum_{b=1}^B (z_{k,(-b)} - \bar{z}_{k,(\cdot)}) (z_{k,(-b)} - \bar{z}_{k,(\cdot)})^T. \quad (10)$$

Hartlap-type correction. An inverted sample covariance overstates precision; for n independent Gaussian samples of dimension p , $E[\hat{\Sigma}^{-1}] = \frac{n-1}{n-p-2} \Sigma^{-1}$, so the precision is scaled by $h = (n-p-2)/(n-1)$ (here $p = 6$, $h = 0.767$ for 31 blocks). Jackknife replicates are neither independent nor Gaussian, so this is a Hartlap-*type* correction whose adequacy is judged only by independent-record coverage.

Frequency pooling. $\hat{\Sigma}_k \leftarrow \text{mean of } \hat{\Sigma}_{k-2}, \dots, \hat{\Sigma}_{k+2}$ (fewer at the band edges), exploiting that the covariance varies smoothly with ω ; the per-bin Hartlap-type factor uses the pooled degrees of freedom (interior $h = 0.953$).

A single record can report honest uncertainties if its covariance is regularised across frequency. A raw 6×6 jackknife covariance from 31 blocks is far too noisy to use as weights (95% coverage 0.705, reported σ too small by 45 to 60%); the Hartlap-type factor alone only reaches 0.78 to 0.79. Pooling over ± 2 bins restores 0.925 to 0.930, and the pooled estimator's scatter equals the oracle-weighted scatter (1.00 to 1.02): pooling regularises the weights without degrading the estimate. At A the pooled scatter/reported ratio is 1.13, but the oracle itself is 1.11 there, so this is not a jackknife failure: the record-derived uncertainty is calibrated to the oracle's level.

Pooling caveats. The ± 2 -bin width was fixed on the earlier reduced-band dry run (different records), before the full validation, and was not tuned on it. Pooling assumes the covariance is smooth over ± 0.12 rad/s; near sharp spectral or SNR transitions (the JONSWAP low-frequency edge, a narrow or bimodal sea) it can bias the covariance.

6.5 G3c: transporting the nominal covariance

Using $\Sigma(p_0)$ at the truth point gave emp/claimed about 1.1 at both points and 95% coverage 0.950 (A) and 0.880 (B). Transporting a nominal covariance through parameter space can be optimistic, which supports record-adaptive uncertainty (G3b); with one independent 200-record calibration per point the A versus B difference is supporting evidence only.

6.6 Residuals and interpolation

Final $\chi^2/\text{dof} = 1.093$ (A) and 1.025 (B) against a standard deviation of about 0.063 for 502 degrees of freedom: the fits are statistically consistent. The node-interpolated model differs from the direct EMM by at most 1.7×10^{-5} in T , checked independently at both truth points, with $d_{\text{interp}} \approx 0$; the 0.2 rad/s model grid is frozen.

7 Major limitation: inverse crime

The synthetic records and the inverse model use the same EMM physics. Different node spacings (0.1 for the records, 0.2 for the solver) change the interpolation, not the physics. The χ^2/dof near 1 is therefore evidence of statistical consistency, not of model adequacy. G3 validates the estimator, the finite-record covariance, the bias model, the solver and the record-derived uncertainty; it does *not* validate the EMM for a real floe. Assumptions still shared by simulation and inversion:

- one coherent incident field from a known direction (no directional spreading, no multiple independent sources);
- one JONSWAP sea state ($H_s = 0.05$ m, $\omega_p = 5$ rad/s, $\gamma_J = 3.3$);
- white, mutually independent sensor noise with an exactly known PSD for the reference correction;
- γ fixed at its true value;
- the Level 2C four-sensor geometry, placed exactly, perfectly synchronised and aligned;
- the EMM truncation [50, 10, 10] and physics, and a circular floe in water of known depth.

8 Files and outputs

File (.../SSI Twin/)	Role	Outcome
runTransmissibilityCovariance.m	U1: covariance treatments A to D, held-out calibration, bias projection	$d_{\text{bias}} \approx 0.9$ at $L = 1024$
runL2048BiasVarianceCheck.m	Segment-length refinement, no EMM solves	F selected
runTransmissibilityTwinInversion.m	U2 / G3: nonlinear inversion, G3a/b/c coverage	all gates pass

Table 6: Sunday’s files, in Inverse Mapping/Parameter Inversion/SSI Twin/ (to be renamed Output-Only Twin/); results and caches in Inverse Mapping/Parameter Inversion/Results/.

Weekend outcome. The chain EMM physics $\rightarrow$ output-only transmissibility $\rightarrow$ finite-record Welch estimator $\rightarrow$ validated covariance and resolution bias $\rightarrow$ nonlinear recovery of (β, R) is complete in the twin, with a field-style uncertainty estimate formed from the record itself. For one 109-minute record at 10% sensor noise: β to about 0.5% and R to about 0.09% (1σ), with a predicted and observed systematic offset of about +0.1% and +0.01%. U2/G3 is complete; the spectrum-prior study (U3) is deferred.

9 Next steps: from synthetic validation to the measurement chain

The project now moves from inverse validation to mission parameterisation and measurement-chain characterisation. Every item below either removes one of the shared assumptions of Section 7 or supplies a number the frozen configuration needs.

9.1 Floe characterisation: field-testable R , β , γ

- **R :** measure the floe (or model floe) diameter directly (tape, drone or overhead image); it is also the parameter the method recovers best, so it is the cleanest independent check.
- **$\gamma = d/H$:** measure thickness h and density; draught $d = \rho_i h / \rho_w$; confirm the water depth H . This is the input that justifies fixing γ (supervisor to confirm on Monday).
- **$\beta = D/(\rho_w g H^4)$:** $D = E h^3 / (12(1 - \nu^2))$ requires E (and ν); obtain an independent estimate from a static loading or bending test, or material data for the model floe, as the ground truth for $\hat{\beta}$. Note the h^3 sensitivity: 1% in h is 3% in β , six times the method's 1σ .
- Rerun the forward-model convergence check (truncation) for the actual (β, γ, R) and frequency band.

9.2 IMU noise characterisation

- Static (and, if possible, floating-still) records per unit: noise PSD in the one-sided per rad/s convention of S2, Allan variance for bias drift; test whether the noise is white over 3 to 8.5 rad/s.
- Channel-to-channel and unit-to-unit noise correlation (the reference correction and the block-diagonal covariance both assume independence).
- Accelerometer scale factor, bias and axis misalignment calibration; convert the measured noise to a realistic replacement for the 10% stress level.
- Twin sensitivity: rerun the corrected estimator with S_{nn} mis-specified by $\pm 30\%$ and with coloured noise, and project the resulting bias with Eq. (4).

9.3 Sensor array configuration and time synchronisation

- **Synchronisation:** a time offset δt between sensor j and the reference multiplies T_j by $e^{-i\omega\delta t}$, a systematic phase ramp. Per-bin phase scatter is of order 10^{-2} rad, so $\omega_{\max}\delta t \ll 10^{-2}$ suggests δt well below about 1 ms; the actual tolerance should be computed by projecting the ramp through J (Eq. (4)), because a systematic error accumulates across all 84 bins. Decide on a shared clock (common trigger, PPS or GNSS time) accordingly.
- **Sampling and filtering:** confirm the usable rate ($\Delta t = 0.1$ s gives Nyquist 31.4 rad/s), anti-alias filtering, and that filter phase is identical across units (unequal filters are another phase ramp).
- **Geometry and orientation:** placement tolerance (project a radial offset through the EMM), sensor axis alignment with the vertical and the incidence axis, coordinate convention.
- **Layout and reference:** keep Level 2C as the baseline; confirm the reference channel (s26, largest response) and check the information for alternative layouts only if the hardware forces a change.
- Achievable record length (battery, storage, stationarity of the sea), which sets N and hence whether $L = 2048$ remains adequate.

9.4 Experiment and remaining computational checks

- **Experiment:** tank or field trial with the characterised array; if possible a wave probe for S_u (Scenario W), which also enables the deferred spectrum-prior analysis (U3).
- **Robustness in the twin** (cheap, reuses the frozen pipeline): directional spreading or a second independent incident direction; model error in the inversion (slightly wrong γ , different truncation); sensor placement and synchronisation errors; a different sea state (narrower JONSWAP, bimodal) with the segment-length rule rechecked.
- Rename SSI Twin/ to Output-Only Twin/; write the weekend chain into the report.

Monday supervisor points. Fixing γ from measured thickness; SSI as a comparison branch only; availability of a wave probe (Scenario W); IMU model, noise specification and synchronisation options; how E (hence β) will be measured independently for the test floe.